

CHRISTOPHER HALL

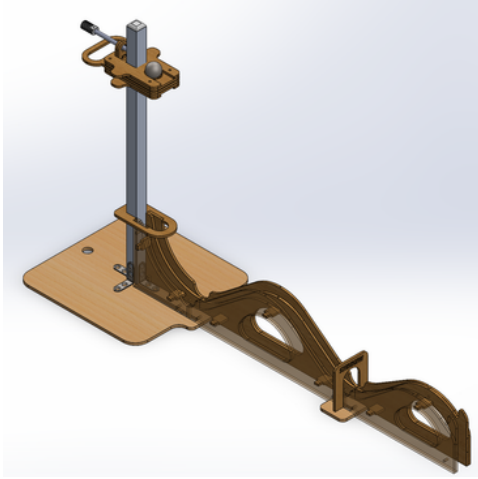
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Energy Conversion Children's Game

February 2026 - April 2026

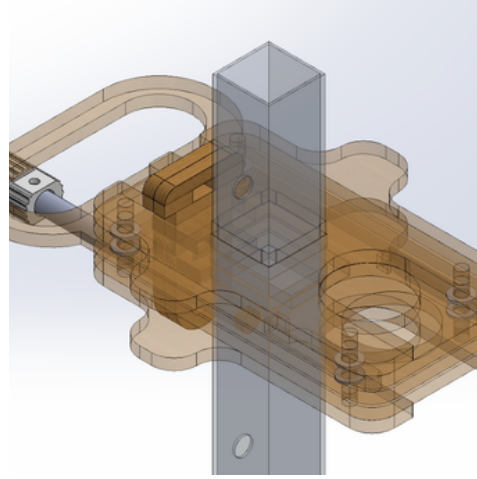
Year 1 Engineering Design Project

University of Southampton Malaysia



What?

- to design a child-friendly, inclusive, educational, sustainable, fun **physical attraction**.
- Demonstrating gravitational potential to kinetic energy conversion for **users aged 7 to 12**.
- To predict the performance of the product and compare it with real-world performance.



How?

- Designed using **Solidworks**.
- Made of laser-cut wood and off-the-shelf components including a hidden spring loaded ball release mechanism.
- Optimised laser-cut components to reduce waste by **20%**.
- Created a custom **python model** to predict heights of track slopes.
- Analysed high-speed video to calculate energy losses and adjust slope height.



Results

- The product is intuitive, easy to use.
- Fun as it is based on an arcade game: bowler roller.
- Cost less than **\$30** to produce due to optimised material and components.